

Ahmad Mohammad

MECHATRONICS / MECHANICAL ENGINEERING — BENG, DTU

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SHEET 1/1

SCOPE MECHATRONICS ENG.

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PORTFOLIO: A22BST.GITHUB.IO

EDUCATION

BEng Mechanical Engineering — Mechatronics & Robotics

Technical University of Denmark (DTU)

Bachelor thesis: Design and Evaluation of Terrain-Adaptive Landing Gear for a Holybro X650 drone (2-person project)

- Self-levelling drone landing gear: three motorised telescopic legs, each stopping itself on ground contact. My part: mechatronics and system integration — built and wired the electronics, wrote the ESP32 firmware, designed a custom KiCad controller board, built a PX4/Gazebo simulation to compare adaptive and rigid landings, and researched the mechanism principles. Team physical tests settled at 2.1–2.9° tilt, incl. a 30° slope with a 160 mm block.

EXPERIENCE

Mechatronics Engineer Intern

2025.09–2026.01

COBOD International A/S — R&D, 3D concrete printers

- Built a PLC-based pressure and flow monitoring system for the printers' additive-dosing line on an Omron NX102 with a remote EtherNet/IP rack: selected and integrated an IO-Link pressure sensor and a turbine flow sensor, and programmed a physics-based flow model with dual independent flow measurement in structured text.
- Designed and ran a systematic flow test protocol that showed the pump under-delivering its nominal flow; coded a PI pump-speed controller and auto-calibration (not validated in operation because of pump faults) and handed over with a detailed report.
- Developed a retrofittable, electronics-free cleaning concept for the printers' Z-axis racks: supplier work, bristle-material selection, cost trade-offs, 3D-printed test prototypes and a Fusion 360 CAD proof of concept.
- Assembled a printer control box and Z cartridge, validated a new PCB, and helped coordinate the company's relocation.

Volunteer Member (Engineering)

2025.06–PRESENT

Engineers Without Borders (Denmark)

- Contributing to early project scoping and planning for rehabilitating water towers: defining requirements, constraints/risks, and an execution plan.

SKILLS

EMBEDDED ESP32 · Arduino · KiCad PCB design · Node-RED · sensor integration (IO-Link, ToF, encoders)

CONTROLS Omron PLC (Sysmac Studio, ST, ladder) · IO-Link · EtherNet/IP · PI/PID control · TwinCAT 3

ROBOTICS / UAV PX4/Pixhawk · MAVLink · Gazebo · UR10 · MiR100 · Festo MPS

DESIGN SolidWorks · Fusion 360 · 3D-print prototyping · mechanism research · materials selection

PROGRAMMING Python · MATLAB/Simulink · C++ (Arduino/ESP32)

CERTIFICATIONS ISO 21500 & 21502 · P-17 welding/thermal cutting

LANGUAGES Danish (native) · English (proficient) · Arabic (proficient)

OTHER Driver's licence · Available immediately

SELECTED PROJECTS

- Compost Buddy** [ACADEMIC, TEAM OF 8] — Smart compost unit with sensors, a motorised stirring arm and an IoT dashboard; I worked on electronics and sensor selection, the mixing mechanism, water handling, CAD and the bill of materials.
- Foosball Robot** [ACADEMIC, TEAM OF 2] — Camera-tracked goalkeeper on stepper-driven timing belts; I wrote the requirements, ran the speed experiments and did the mechanical design and CAD.
- AMR-Cobot Integration** [ACADEMIC, TEAM OF 5] — MiR100 zones and missions with error handling, and a UR10 force-based quality check on lid mounting.
- Angle-Bracket Fabrication** [ACADEMIC, TEAM OF 4] — Calibrated a press's force measurement against a proving ring; compared calculated and measured punching and bending forces and spring-back.